

Harsh Mishra

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Professional Summary

Results-driven AI/ML Engineer with hands-on experience designing NLP chatbots, RAG architectures, and computer vision models. Skilled in deploying scalable, end-to-end AI solutions using Python, LLMs, and FastAPI.

Skills

Programming Languages: Python

Backend & Frameworks: FastAPI, Flask, Django

Machine Learning & AI: Machine Learning, Deep Learning, NLP, Computer Vision, scikit-learn, OpenCV

Databases & Data Tools: MySQL, SQLite, NumPy, Pandas

Tools & Platforms: Git, GitHub, Docker, Jupyter Notebook, Streamlit

Professional Skills: Problem Solving, Debugging, Documentation, Team Collaboration

Education

Master of Computer Applications

2024 – 2026

Guru Ghasidas Vishwavidyalaya, Bilaspur, C.G — CGPA: 8.00

Bachelor of Computer Science (Hons)

2021 – 2024

Guru Ghasidas Vishwavidyalaya, Bilaspur, C.G — CGPA: 7.78

Experience

Tech Intern — CGPD TM, Delhi

Apr 2026 – Sep 2026

AI/ML Development, Backend Development

- **Generative AI & Chatbot Development:** Architected IP Saarthi 2.0, an NLP-powered conversational chatbot built using Python, LLMs, and Prompt Engineering to streamline intellectual property inquiries.
- **RAG-Powered Text-to-SQL Agent:** Developed an autonomous Text-to-SQL AI Agent on SQL Server Database, automating schema retrieval, dynamic SQL query generation, tabular reporting, and chart-based data visualizations.
- **Document Intelligence & OCR:** Engineered an end-to-end RAG pipeline by scraping IP legal documents, integrating OCR, and fine-tuning document chunking and parsing strategies to enhance context retrieval accuracy.
- **Query & Performance Optimization:** Optimized LLM prompt workflows, query validation, and error handling mechanisms to improve generation accuracy and system response times.
- **Database Pipeline Automation:** Built automated Python and SQL scripts for high-volume Trademark BLOB file extraction and data preprocessing.

Projects

Multi-Agent SQL AI System

Jan 2026 – March 2026

- Developed a multilingual, RAG-powered Natural Language to SQL system using FastAPI and Ollama for secure, schema-aware query generation. *Python, FastAPI, Ollama, FAISS, SQLite, RAG*

Image Captioning System with Object Detection

Aug 2025 – Dec 2025

- Designed an encoder-decoder image captioning model integrating YOLOv8 and VGG16 for highly accurate, context-aware semantic descriptions. *Python, Deep Learning, Computer Vision*

Weather Prediction Project

June 2025 – July 2025

- Built and deployed a Streamlit-based machine learning regression model trained on 100,000+ historical records to predict climate metrics. *Python, Machine Learning, scikit-learn*

Certificates

Machine Learning for Data Science Projects — IBM Skills Build

Feb 20, 2025

Learn Python & Ethical Hacking From Scratch — Udemy

July 17, 2025

Exploring Data Transformation with Google Cloud — Google Cloud Training

Oct 3, 2024